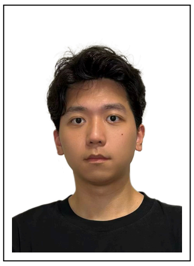
Wang YuHeng

Data Science Undergraduate

Shenzhen
China
+86 137 1424 1526
ywangrg@connect.ust.hk



Professional Summary

Third-year Data Science student at HKUST with strong foundation in machine learning, computer vision, statistical modeling, and programming. I have since applied my Python and machine learning skills to train a U-Net structure for heart detection and segmentation that achieves accuracy over 90% and F1-score 0.7992. Have done research with Gary Chan on 3D-reconstruction by utilizing 2D floorplan to improve efficiency.Self-studied financial data analysis and have basic accurary knowledge. Seeking opportunities in data analyst, AI/CV research.

Education

09/2023 – 09/2027 (expected) **B.Sc. in Data Science**, *The Hong Kong University of Science and Technology (HKUST)*, Hong Kong

GPA 3.733 / 4.3 (Major GPA: 3.834)

Ranking Top 10%

Relevant Coursework
- **Mathematics & Statistics**: Honours Calculus I/II [A], Multivariable Calculus [A+], Statistics [A], Honor Probability [A+], Mathematical Analysis [A], Matrix Computation [A], Matrix Differentiation[A], Bayesian Statistics[A], Statistical Inference[A]
- **Computer Science & ML**: C++ & Data Structures [A], Discrete Mathematics [A+], Machine Learning [A], Algorithms [B+]

Research Experience

Sep 2025 – Dec 2025 **Undergraduate Researcher – 3D Point Cloud Reconstruction**, *HKUST UROP / Course Project*, Hong Kong

Developed floorplan-guided framework to align/fuse multiple point clouds using 2D floorplans as structural priors, improving geometric consistency and scale normalization for indoor 3D scenes (digital twins, robotics, BIM).

Summer 2025 **Research Intern – Medical Image Segmentation**, *Seoul National University (SNU-HKUST Sponsored Project)*, Seoul, Korea

Built and trained U-Net model for automatic heart detection, location prediction, and segmentation in X-ray images and achieved 90% accuracy and 0.7992 F1 score.

2026–Present **FYP – Towards a Personal Embodied AI Agent with Reinforcement Learning**

Developing a perception-action pipeline using PyTorch and Reinforcement Learning to enable embodied agents to execute visual search and manipulation tasks based on natural language instructions.

Additional Projects & Self-Study

Oct 2024 **Machine Learning Project**

Built a clustering-based CAPTCHAs model by utilizing pure numpy model myself which is applicable for human authorization

Dec 2024 **Machine Learning Project**

Applying MLP to build a Spam Classification model which achieves 95% accuracy and 0.045 loss in binary-cross-entropy loss

May 2025 **C++ Project**

Built a data structure to mimic Excel behavior using BST and hashtable which can be used to effectively storage data

Summer 2025 **Financial Data Analysis & Quantitative Modeling (Self-Study)**

Mastered Pandas for financial data processing, time-series cleaning, and analysis.

Implemented Markowitz Mean-Variance Model and Black-Litterman (BL) Model for portfolio optimization and asset allocation.

Aug 2026 **Data Visualization Project**

Visualization topic: Visualization of AI's impact on job market. (`https://github.com/TheLogarhythm/Career-a-vis-AI`)

Mastered various data visualization tools including Datawrapper, Tableau, pandas, iNZight, and RAWGraphs

Skills

Programming Languages: Python(Favorite) , C++ , Java, R

Languages: Mandarin(Native Speaker), English